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Cardiovascular risk prediction via ensemble machine learning and oversampling methods

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Cardiovascular diseases are a leading cause of global mortality, with hypertension, obesity, and other factors contributing significantly to risk. Artificial Intelligence has emerged as a valuable tool for early detection, offering predictive models that outperform traditional methods. This study analyzed a dataset of 709 individuals from Ecuador, including demographic and clinical variables, to estimate cardiovascular risk. During preprocessing, records with missing values and duplicates were removed, and highly correlated variables were excluded to reduce multicollinearity and prevent overfitting. The performance of several machine learning algorithms–including Decision Trees, Random Forest, Gradient Boosting, Extreme Gradient Boosting, LightGBM, Extra Trees, AdaBoost, and Bagging–was compared, while addressing class imbalance using SMOTE and a hybrid ROS–SMOTE approach. Gradient Boosting with the hybrid technique achieved the best performance, obtaining an accuracy of 0.87, a precision of 0.81, a recall of 0.74, and an F1-score of 0.75. Its superior performance is attributed to its sequential error correction mechanism and integrated regularization strategies, which effectively reduce overfitting and improve generalization in noisy or imbalanced datasets. These findings demonstrate the potential of AI-based models to improve early detection and management of cardiovascular disease, highlighting the importance of anthropometric, clinical, and blood pressure variables in predicting cardiovascular risk.

Keywords Cardiovascular disease, Machine learning, Classification, Balanced techniques

Cardiovascular disease (CVD) remains one of the leading causes of mortality worldwide. The cardiovascular system comprises the heart and its blood vessels. CVD encompasses four major conditions: coronary artery disease, cerebrovascular disease, peripheral artery disease, and aortic atherosclerosis[1], which can lead to myocardial infarction, cardiac arrhythmias, or stroke[2].

Risk factors for CVD include unhealthful dietary patterns, physical inactivity, dyslipidemia, prediabetes/diabetes, hypertension, obesity, age, race/ethnicity, sex differences, thrombosis, smoking, kidney dysfunction, and genetics/familial hypercholesterolemia[3]. The prevalence of CVD and its associated risk factors is rising in low- and middle-income countries[4], which account for more than 80% of all CVD-related deaths worldwide[5].

Clinical recommendations for the prevention of cardiovascular disease often advocate the use of prediction models or risk scores, which evaluate the combined impact of multiple quantified risk factors to identify individuals most likely to benefit from preventive measures and to estimate future disease risk[6]. Over the past five decades, several cardiovascular disease risk scores have been developed and updated, including the Framingham Risk Score, Pooled Cohort Equations (PCE), Systematic Coronary Risk Evaluation (SCORE2), WHO cardiovascular disease risk charts, PREDICT, QRISK3, and Predicting Risk of Cardiovascular Disease Events (PREVENT)[6].

With advances in artificial intelligence, several researchers have applied machine learning algorithms to develop models for predicting the risk of heart attacks. One study[7] used a dataset of 8,763 patients obtained from Kaggle and compared multiple machine learning methods, reporting the best performance with Support Vector Machine (SVM), Logistic Regression, K-nearest Neighbor (KNN), and Random Forest (RF). Similarly, another study[8] employed various machine learning techniques to predict the same disease using a dataset of 301 patient records that included demographic and health-related characteristics such as gender and health status. Algorithms applied in this study included RF, regression models (RM), KNN, Naive Bayes (NB), SVM, Decision Trees (DT), Extreme Gradient Boosting (XGBoost), and neural networks (NN). Among these, RF achieved the highest accuracy of 0.885.

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Several studies have investigated the application of machine learning techniques for the diagnosis and prediction of cardiovascular disease (CVD). In one study[9], the authors evaluated the effectiveness of eight machine learning-based imputation methods to handle missing data in predictive models. The dataset comprised 10,264 records of the southern Xinjiang population, China. The evaluated imputation techniques included Simple imputation, Regression imputation, Expectation maximization (EM), Multiple imputation (MICE), KNN, Clustering imputation (CI), RF and DT. In addition, a model was built to predict cardiovascular risk using SVM. The results indicated that KNN and RF demonstrated superior performance in data imputation tasks. Another study[10], machine learning algorithms such as DT, RF, SVM, Logistic Regression (LR), and KNN were applied to a dataset collected from a family health unit in Turkey in 2018, excluding individuals under the age of 40. Based on precision and recall metrics, KNN exhibited the highest accuracy in identifying low-risk patients. while DT and RF also yielded strong classification performance. A further study[11] analyzed data from 1508 individuals residing in China using LR, SVM, DT, RF, KNN, Naive Bayes (NB), and XGBoost. The dataset included demographic information, medical history, cytokine profiles, and other clinical indicators relevant to CVD. The results showed that SVM and LR provided the best predictive performance.

A study[12] performed a comparative analysis of algorithms including SVM, KNN, NB, DT, RF, and artificial neural networks (ANN) for CVD prediction, with RF achieving the highest accuracy of 0.92. Subsequent research[13] combined fuzzy logic with knowledge-based ANN models, reaching an accuracy of 0.95 and a precision of 0.978. This work utilized the Cleveland Heart Disease dataset from the UCI repository, comprising 303 records and 13 selected attributes, including the target variable. The methodology was later extended to three additional datasets, incorporating various machine learning and deep learning algorithms. Feature selection was performed using attribute importance scores, ranking methods, and established selection algorithms. To enhance classification performance, a Positional Weighted Feature Matrix (PWFM) was developed, aggregating the posterior probabilities of multiple base classifiers within a heterogeneous ensemble voting framework. The final diagnostic output was obtained by integrating these results through multiple ensemble techniques[14].

Some authors highlight the benefits of tree-based algorithms. These algorithms are valuable tools for classification and prediction tasks, particularly in the health field, as they allow data to be divided into simple subsets organized in a hierarchical structure of rules, which facilitates interpretation[15]. They note that ensemble methods, such as RF, have been developed to mitigate overfitting by combining multiple DTs trained on random subsets of the data, thereby improving accuracy and reducing variance. Another approach discussed is Gradient Boosting, which builds sequential models by correcting errors from previous predictions, achieving higher accuracy, albeit with a greater computational cost.

Decision tree learning is the process of constructing such a tree from examples of inputs and their corresponding outputs. DT presents hierarchically structured information that facilitates interpretation[16]. The use of RF is highlighted for combining multiple DTs and reducing overfitting by aggregating their predictions. Gradient Boosting (GB) is described as a technique that sequentially refines models by correcting errors with each new prediction. Regarding Extra Trees, this algorithm introduces an additional level of randomization in node splitting, differentiating itself from Random Forest by generating more varied cuts, which can increase model stability. Similarly, AdaBoost is presented as an iterative technique that adjusts the weight of each observation, emphasizing those that are most difficult to classify to enhance overall model performance. Finally, bagging is described as a strategy that trains multiple models on random subsets of the data using sampling with replacement, reducing variance and improving model robustness.

Previous studies in the prediction of cardiovascular disease (CVD) have several limitations. Firstly, the datasets used are synthetic or collected outside of Latin America and do not reflect the Ecuadorian reality of patients with cardiovascular disease, particularly in terms of demographic data and lifestyle. Secondly, most of these datasets are imbalanced, and the methodologies proposed in the literature typically preprocess the data without applying oversampling techniques to balance the minority classes during the training phase. Finally, although various machine learning techniques have been tested, such as SVM, Random Forest, and XGBoost, comparative studies that systematically focus on tree-based algorithms remain limited.

To address these gaps, this study proposes a novel framework for CVD risk prediction using real Ecuadorian clinical data, representing one of the first region-specific contributions in this field. Our approach involves a multiclass classification task with eight tree-based algorithms (Decision Trees, Random Forest, Gradient Boosting, XGBoost, LightGBM, Extra Trees, AdaBoost, and Bagging), each optimized through Grid Search for robust evaluation. Furthermore, we employ two balancing strategies: SMOTE and a hybrid method combining Random Oversampling (ROS) with SMOTE. By integrating localized data, a multiclass prediction framework, and balancing techniques, this work contributes new insights to CVD predictive modeling, particularly within underrepresented Latin American populations.

The remainder of the paper is organized as follows: Section "Methodology" presents the methodology of the proposed research. Section "Results" details the experiments, including two data balancing techniques and eight classification algorithms. Section "Discusion" provides a discussion of the results and their implications. Finally, Sect. "Conclusion" concludes the paper and outlines directions for future work.

Methodology

This study was carried out following a process organized into several stages, including data collection, processing, dataset partitioning, class balancing, model training with GridSearch, and performance evaluation. These stages for predicting cardiovascular risk are illustrated in Fig. 1.

Several Python libraries, including pandas, scikit-learn, and imblearn.over_sampling, were employed in some of these stages.

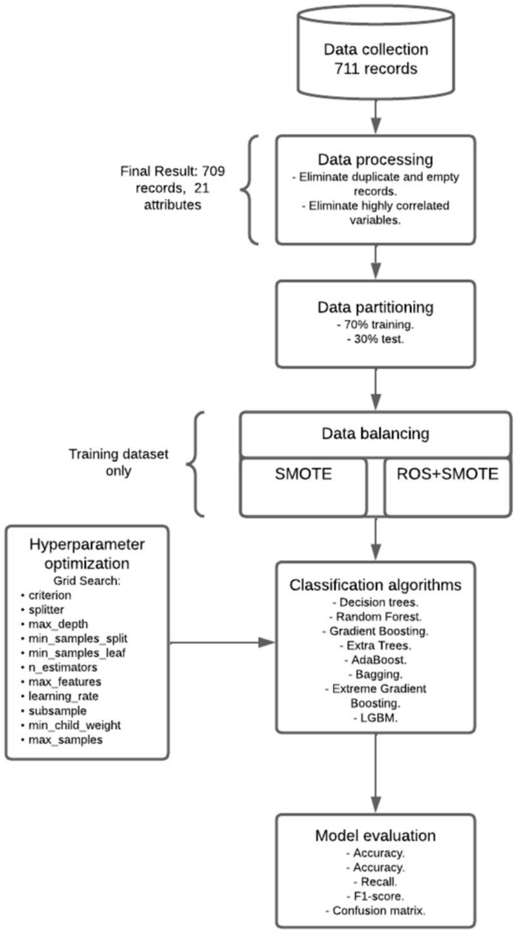

Fig. 1. Stages of methodology.

Data collection

In Ecuador, CVD represents the leading cause of mortality, according to data from the 2018 STEPS surveys (PAHO, 2023). The data were obtained from a quasi-experimental study conducted in Loja, Ecuador, between November 2023 and February 2024. A total of 1,388 workers from various private and public institutions affiliated with the Ecuadorian Institute of Social Security (IESS) Health Center were initially screened for eligibility. Of these, 711 individuals met the inclusion criteria: adults aged 40 to 75 years who were assessed with a Globorisk score to calculate their 10-year cardiovascular risk. Exclusion criteria included a prior diagnosis of cardiovascular disease, cognitive impairment, and pregnancy. The dataset included demographic and clinical attributes. Details of the data collected are presented in Tandazo et al.[17].

Ethical approval for this study was obtained from the Institutional Ethics Committee of the Hospital San Francisco, Quito, Ecuador (Institutional Review Board approval code 031). Written informed consent was obtained from all participants. Data confidentiality was safeguarded through anonymization procedures. This study was carried out taking into consideration the ethical principles of the Declaration of Helsinki.

Data preprocessing

Preprocessing techniques were used to prepare the data before applying the classification algorithms. Records with empty values and duplicates were eliminated. Some variables were observed to be highly correlated with each other, which could introduce redundancy and negatively affect the performance of the models. To avoid multicollinearity and overfitting problems, it was decided to eliminate the following characteristics: weight, height, index body mass, muscle mass, sedentary lifestyle. At the end of this step, the dataset consists of 709 records and 21 attributes: sex, age, level of instruction, marital status, work activity, monthly income, smoker, former smoker, alcohol consumption, alcohol consumption in the last 30 days, number of MET per week, weekly level of physical activity, adherence to the Mediterranean diet, abdominal circumference, systolic blood pressure, diastolic blood pressure, waist-hip index, body fat percentage, body fat mass, visceral fat level, muscle mass index. See Table 1 for more details.

The variable to classify *RIE_CARD_A* has four values: 0 for low cardiovascular risk, 1 for moderate cardiovascular risk, 2 high cardiovascular risk, and 3 for very high cardiovascular risk.

Num	Variable name	Description
1	EDAD	Age
2	SEXO	Sex
3	N_INSTRUC	Level of education
4	EST_CIV	Marital status
5	ACTV_LAB	Work activity
6	INGR_MENS	Monthly income
7	FUMA	Smoker
8	EX_FUM	Former smoker
9	CONSUM_ALCOH	Have you ever consumed alcohol?
10	CONSUM_ALCOH_30	Have you consumed alcohol in the last 30 days?
11	METS	Number of mets/week.
12	NIVELDEACTIVIDADSEMANAL	Weekly physical activity level.
13	INTERPRETACIóN	Adherence to Mediterranean diet.
14	PER_ABD	Abdominal circumference.
15	TASist	Systolic blood pressure.
16	TADiast	Diastolic blood pressure.
17	CINT_CAD	Waist-hip index.
18	PGcorp	Body fat percentage
19	MGcorp	Body fat mass
20	NGVisc	Visceral fat level
21	MMEIMC	Muscle mass index
22	RIE_CARD_A	Cardiovascular risk level

Table 1. Description of variables.

Class	100% data	70% data Train	30% data Test
0	463	324	139
1	179	125	54
2	59	41	18
3	8	6	2
Total	709	496	213

Table 2. Data partition.

Data partitioning and balancing

The data were divided into 70% for training and 30% for testing (see Table 2).

In this work, two different data balancing techniques were employed on the training data:

1. SMOTE only: In the first approach, we applied SMOTE directly to the training set. SMOTE generates synthetic samples by selecting each minority-class instance and creating new examples along the line segments connecting it with its k nearest minority-class neighbors[18]. This method helps to reduce class imbalance by producing high-quality synthetic data for underrepresented classes[19].
2. Hybrid strategy (ROS + SMOTE): In the second approach, we implemented a two-step hybrid balancing process. First, ROS was applied to increase the frequencies of classes 2 (high risk) and 3 (very high risk) to match that of class 1 (moderate risk). ROS randomly duplicates minority-class samples to balance the dataset[20]. This step ensured that there were enough real examples for minority classes before generating synthetic data, thereby reducing the risk of noise. Next, SMOTE was applied to the ROS-adjusted data to generate synthetic instances for classes 1, 2, and 3, balancing them with class 0 (low risk). As a result, all four classes achieved equivalent representation, ensuring that minority classes were sufficiently reinforced prior to interpolation and that the overall distribution was balanced for model training.

Classification algorithms

For the classification task, eight algorithms were applied: Decision Trees (DT), Random Forest (RF), Gradient Boosting (GB), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM),Extra Trees (ET), AdaBoost (AB) and Bagging (BG).

These methods were selected because they improve both performance and robustness by combining multiple learners. Ensembles of decision trees reduce individual errors, such as overfitting to the training data, and enable the detection of complex patterns that a single model may fail to capture. For example, a study[15] proposed

Class	Imbalanced data	Balanced data
0	324	324
1	125	324
2	41	324
3	6	324
Total	496	1296

Table 3. Balancing with SMOTE.

Class	Imbalanced data	ROS balancing	SMOTE balancing
0.0	324	324	324
1.0	125	125	324
2.0	41	125	324
3.0	6	125	324
Total	496	699	1296

Table 4. Balancing with ROS-SMOTE.

HeartEnsembleNet, a hybrid ensemble learning framework that demonstrated strong efficacy in cardiovascular risk assessment. This work incorporated two ensemble strategies–voting and stacking–to enhance classification performance. Similarly, another work[21] applied ensemble machine learning algorithms to classify patients into low- and high-risk groups, highlighting the importance of ensemble techniques for capturing complex, non-linear interactions among risk factor attributes. An additional observation is that the application of ensemble methods to lifestyle, demographic, and clinical variables for CVD risk assessment remains relatively underexplored.

The eight algorithms worked in a multi-classified process of 4 values for the RIE_CARD_A variable. The Grid Search was used to obtain the best parameters for each algorithm. The selection of hyperparameters vary according to the specific algorithms employed. In this study, most models were configured to allow GridSearch to optimize key parameters, including max_depth, min_samples_split, min_samples_leaf, and max_features. Tuning split and leaf parameters helps avoid overfitting[22]. Furthermore, for certain algorithms, such as GB and AB, the learning_rate was also included in the search space to identify its optimal value. The learning_rate acts as an implicit regularizer, helping to control the complexity of the model and mitigate overfitting[23].

Model evaluation
For evaluation process, accuracy, precision, recall, and F1-score metrics were applied.

Results
Table 2, column 2, shows the number of instances for each class in the dataset (100% of the data), classes 2 and 3 have few samples with 59 and 8 respectively, compared to classes 0 and 1 with 463 and 179 respectively. The balancing process was applied on the training data, Table 2, column 3, shows the values of the instances per class. The results with SMOTE and ROS-SMOTE are shown in Table 3 and 4 respectively.

After training the models using balancing techniques on 70% of the data, the results of testing on the remaining 30% are presented in Table 5. This table shows the evaluation of each algorithm based on four metrics: accuracy, precision, recall, and F1-score. Furthermore, a visual representation of the same results is provided in Fig. 2.

The GB algorithm achieved the best result. With SMOTE, GB achieved an accuracy of 0.84, precision of 0.81, recall of 0.74 and F1-score of 0.75. (See Table 6). These values were improved with the ROS-SMOTE combination, reaching an accuracy of 0.87, a precision of 0.87, a recall value of 0.73 (a little lower compared to the SMOTE experiment) and an F1 score of 0.78. (See Table 7)

The confusion matrix is a useful tool to observe the errors and correct values predicted by the algorithms. The actual and predicted values of the GB algorithm utilizing SMOTE and the hybrid balancing technique are shown in Fig. 3. The GB algorithm with SMOTE correctly classifies 178 out of 213 instances, whereas GB combined with ROS-SMOTE achieves a slightly higher performance, correctly classifying 185 out of 213 instances. These values are obtained by adding the values along the diagonals of each confusion matrix, indicating the number of true positives. This improvement is reflected in the overall accuracy: 0.87 for ROS-SMOTE compared to 0.84 for SMOTE alone (see Tables 5, 6 and 7).

A detailed analysis of the two matrices is given below.

GB with SMOTE (see Fig. 3)

- Class 0 (139 instances): 122 correctly classified; 17 misclassified as Class 1.
- Class 1 (54 instances): 40 correctly classified; 5 misclassified as Class 0, and 9 as Class 2.
- Class 2 (18 instances): 15 correctly classified; 3 misclassified as Class 1.
- Class 3 (2 instances): 1 correctly classified; 1 misclassified as Class 0.

Balance technique	Algorithms	Accuracy	Precision	Recall	F1-score
SMOTE	DT	0.58	0.35	0.38	0.36
	RF	0.76	0.54	0.67	0.58
	GB	**0.84**	**0.81**	**0.74**	**0.75**
	XGBoost	0.82	0.57	0.59	0.58
	LightGBM	0.81	0.54	0.59	0.56
	ET	0.68	0.51	0.61	0.54
	AB	0.72	0.57	0.67	0.58
	BG	0.82	0.54	0.57	0.55
ROS-SMOTE	DT	0.75	0.59	0.6	0.6
	RF	0.78	0.58	0.67	0.61
	GB	**0.87**	**0.87**	**0.73**	**0.78**
	XGBoost	0.85	0.62	0.59	0.60
	LightGBM	0.85	0.65	0.59	0.61
	ET	0.64	0.43	0.55	0.45
	AB	0.71	0.56	0.63	0.55
	BG	0.85	0.88	0.7	0.76

Table 5. Classification results. Significant values are in bold. *DT* Decision Trees, *RF* Random Forest, *GB* Gradient Boosting, *XGBoost* Extreme Gradient Boosting , *LightGBM* Light Gradient Boosting Machine, *ET* Extra Trees, *AD* AdaBoost and *BG* Bagging.

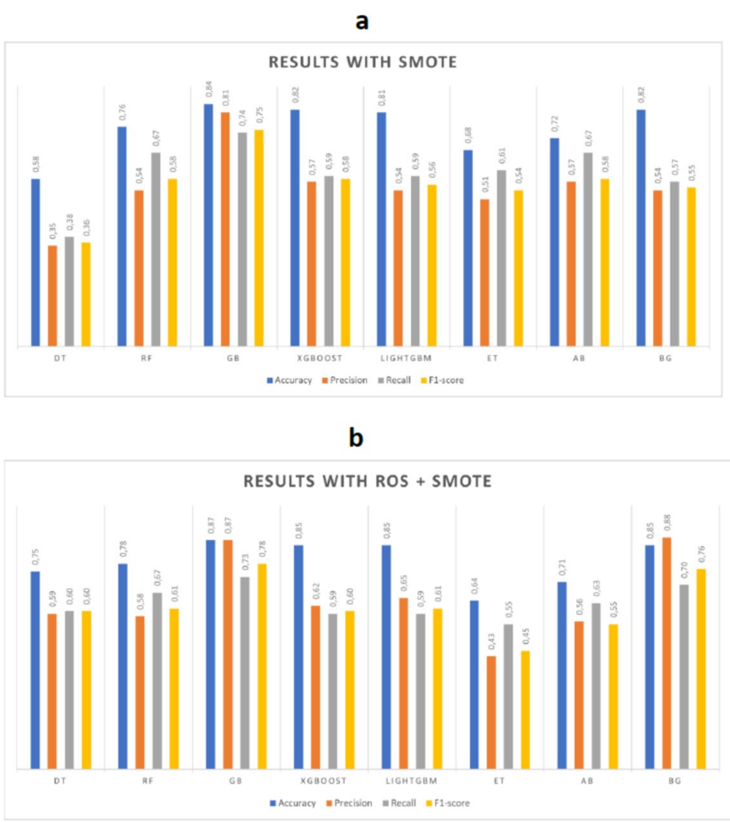

Fig. 2. Visual representation of classification results. (**a**) Represent the results of GB with SMOTE. (**b**) Represent the results of GB with ROS-SMOTE.

GB with ROS-SMOTE (See Fig. 3)

- Class 0 (139 instances): 126 correctly classified; 12 misclassified as Class 1, and 1 as Class 2.
- Class 1 (54 instances): 46 correctly classified; 6 misclassified as Class 0, and 2 as Class 2.
- Class 2 (18 instances): 12 correctly classified; 6 misclassified as Class 1.

Class	Precision	Recall	F1-score	Support
0.0	0.95	0.88	0.91	139
1.0	0.67	0.74	0.70	54
2.0	0.62	0.83	0.71	18
3.0	1.00	0.50	0.67	2
Accuracy			**0.84**	213
Macro avg	**0.81**	**0.74**	**0.75**	213
Weighted avg	0.85	0.84	0.84	213

Table 6. Classification report for GB with SMOTE. Significant values are in bold.

Class	Precision	Recall	F1-score	Support
0.0	0.95	0.91	0.93	139
1.0	0.72	0.85	0.78	54
2.0	0.80	0.67	0.73	18
3.0	1.00	0.50	0.67	2
Accuracy			**0.87**	213
Macro avg	**0.87**	**0.73**	**0.78**	213
Weighted avg	0.88	0.87	0.87	213

Table 7. Classification report for GB with ROS-SMOTE. Significant values are in bold.

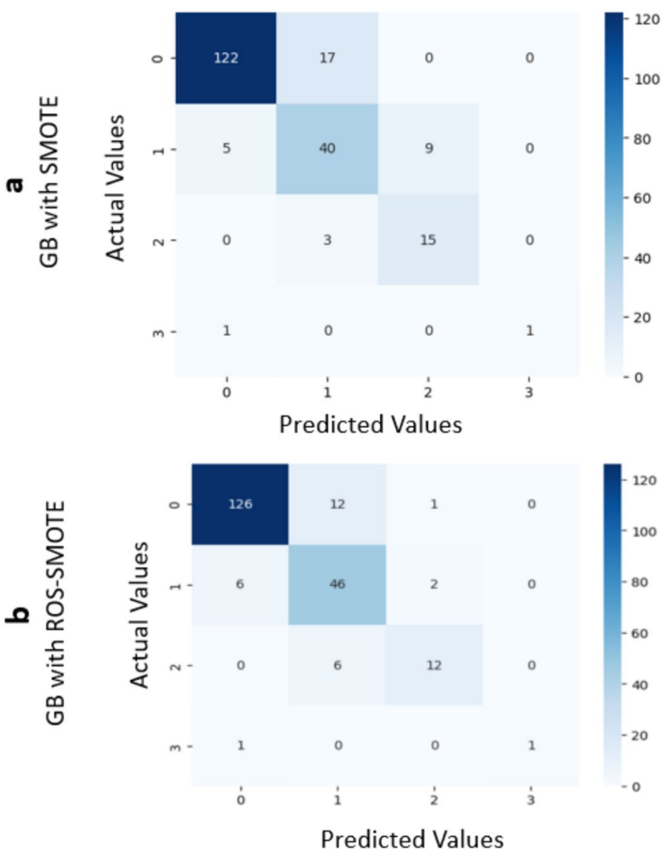

Fig. 3. Confusion matrix. (**a**) Represents the confusion matrix for GB with SMOTE. (**b**) Represents the confusion matrix for GB with ROS-SMOTE.

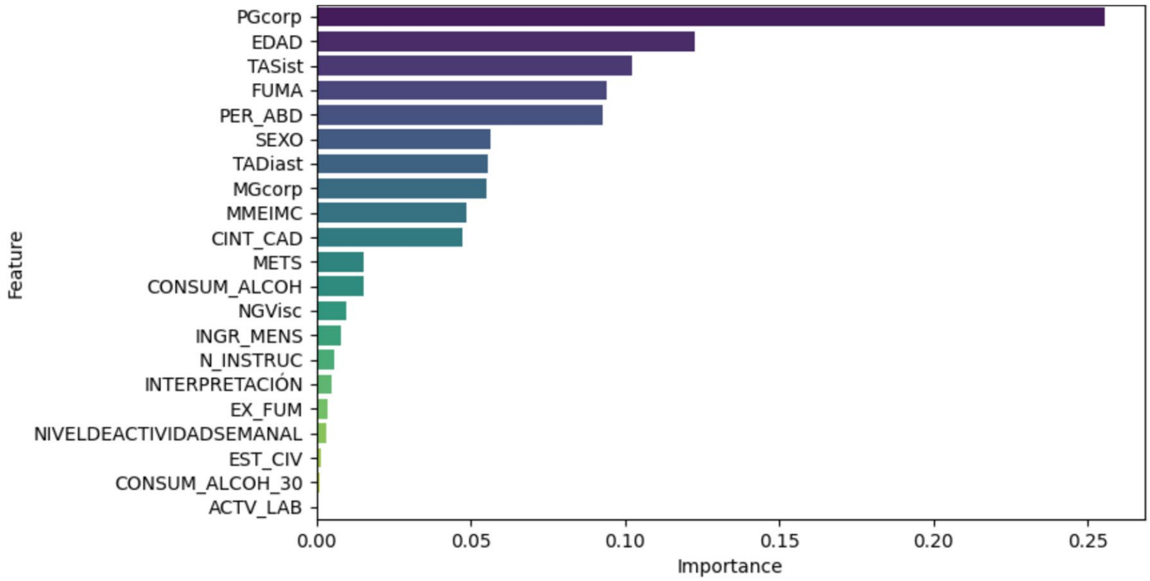

Fig. 4. Feature importance of GB with ROS-SMOTE.

- Class 3 (2 instances): 1 correctly classified; 1 misclassified as Class 0.

In both confusion matrices, zero values indicate that the algorithm did not assign any predictions to the corresponding class.

In general, ROS-SMOTE improves the classification performance for Classes 0 and 1. Class 3 remains unchanged in both techniques. Class 2 shows better performance with SMOTE, achieving 3 more correct predictions. This difference is also reflected in the classification reports: the recall for class 2 is 0.83 for GB with SMOTE (see Table 6) and 0.67 for GB with ROS-SMOTE (see Table 7).

Discusion

This study demonstrates that machine learning models can improve cardiovascular risk prediction, achieving strong outcomes with advanced classification techniques. Eight tree-based algorithms (DT, RF, GB, XGBoost, LightGBM, ET, AB, and BG) were applied in combination with two balancing strategies: SMOTE and a hybrid ROS–SMOTE approach.

A key outcome of this study is the performance achieved by Gradient Boosting (GB) when combined with class balancing, particularly the hybrid ROS–SMOTE strategy, reaching an accuracy of 0.87 and an F1-score of 0.78. The performance of the GB model can be attributed to its ability to sequentially optimize errors made by previous models, making it particularly effective in handling overlapping class distributions–a common challenge in medical datasets where risk categories may share similar clinical features.

Furthermore, GB incorporates regularization techniques such as shrinkage, subsampling, and tree depth control, which help suppress overfitting and improve generalization, especially in noisy or imbalanced datasets. This characteristic is critical in the medical field, where models must generalize well to unseen patient data.

Moreover, Fig. 4 presents the feature importance derived from our best-performing model, the Gradient Boosting (GB) classifier combined with the ROS-SMOTE balancing technique. Several variables contributed substantially to CVD prediction. The top ten features included PGcorp (body fat percentage), EDAD (age), TAsist (systolic blood pressure), FUMA (smoker), PER_ABD (abdominal perimeter), SEXO (sex), TADiast (diastolic blood pressure), MGcorp (body fat mass), MMEIMC (muscle mass index), and CINT_CAD (waist-hip index). These findings indicate that anthropometric and body composition measures play a central role in driving the classification performance. In addition, arterial pressure (both systolic and diastolic) also emerged as important predictors of CVD risk.

For performance comparison, we followed experimental protocols commonly used in the literature, considering preprocessing, dataset partitioning, model selection, and evaluation metrics. However, direct comparison is limited since our dataset–focused on Ecuadorian patients with CVD–is novel, whereas most prior studies rely on Kaggle or proprietary datasets and frame the problem as a binary classification (presence or absence of the disease). Unlike these works, we addressed multiclass prediction and systematically compared eight tree-based algorithms, identifying GB as the most effective.

The literature presents diverse approaches with varying limitations. For example, in Ref.[7] proposed a synthetic dataset built from different countries, which contained 26 variables and two labels: 1 for heart attack risk and 0 otherwise. They used an 80/20 train–test split and experimented with different machine learning methods. After hyperparameter tuning, logistic regression achieved the best accuracy of 64.18%. However, this study presented limitations, as the training data were biased or incomplete, and risk factors may change over time, potentially affecting the model's generalization ability. In Ref.[8], electronic health records and clinical diagnosis reports were used to predict whether a patient is likely to develop a heart attack or is healthy. The dataset included 14 variables. The authors implemented several machine learning algorithms, with Random Forest providing the best performance, reaching an accuracy of 88.52%. Similarly, Ref.[9] handled a dataset with 38 continuous and discrete variables from Xinjiang, China. They addressed missing data by applying eight different imputation methods and then used 10-fold cross-validation on an 80/20 train–test split. The experiments showed that Random Forest achieved the best AUC of 0.777 when trained with the complete dataset, compared to lower performance after imputation (e.g. KNN imputation yielded an AUC of 0.769). In Ref.[10], used a dataset with 21 variables collected from family health units in Turkey in 2018, with the objective of classifying patients into three risk categories: low risk, moderate risk, and high risk. To divide the data, they applied two protocols: K4 (75% train, 25% validation) and K5 (80% train, 20% validation). They implemented five machine learning algorithms and reported results based on the K4 protocol, running it 20 times with different partitions and averaging the validation accuracy. Among the models, KNN outperformed the others, achieving 92% accuracy with K5 and 90% with K4.

Other works explored hybrid or ensemble methods In Ref.[13], the focus was on the early detection of cardiovascular diseases by combining a fuzzy rule-based classification system with an artificial neural network (ANN). The authors used the Cleveland Heart Disease dataset from UCI, applied preprocessing and attribute selection, and divided the data into training (80%) and testing (20%) sets. This hybrid framework achieved strong performance, with an accuracy of 0.952, specificity of 0.987, precision of 0.978, and sensitivity of 0.989. Similarly, in Ref.[15] the authors proposed a hybrid ensemble learning model that integrates multiple machine learning (ML) classifiers for cardiovascular disease (CVD) risk prediction. They used a dataset consisting of patients with and without cardiac disease; however, a key limitation was the absence of electrocardiography data, echocardiography, and biomarkers such as Troponins and NT-ProBNP. After preprocessing, the dataset was split into 80% training and 20% testing. The authors compared their proposed model, Heart EnsembleNet, with Hybrid Random Forest Linear Models (HRFLM) and other ensemble techniques, including stacking and voting. Their model outperformed the alternatives, achieving an accuracy of 92.95

Limitations and ethical implications

The dataset utilized in this study includes demographic and clinical information from 1,388 workers in Ecuador. Data collection procedures adhered to established ethical standards, with informed consent obtained from all participants and institutional approval secured, as outlined in the Data Collection section. However, the study is subject to certain limitations. The relatively modest sample size may constrain the generalizability of the findings to broader populations. Furthermore, reliance on self-reported data introduces the potential for response bias, which could affect the accuracy and reliability of the reported outcomes.

On the other hand, this dataset is novel because it includes four levels of cardiovascular risk,sk, in contrast to the binary classification commonly found in the literature. The main challenge lies in the imbalanced class distribution, which reflects the real-world characteristics of patients in Loja, Ecuador. To address this issue, oversampling techniques were applied to increase the number of samples in the minority classes using synthetic data. However, in future work, additional patient data will be collected in the coming years to strengthen minority classes with real data.

As mentioned in the article, future work will focus on applying LIME and SHAP methods from the outset to interpret the eight tree-based algorithms. Although this is a computationally expensive task, it is essential for obtaining reliable and consistent explanations. In other words, by applying these methods, we aim to enhance the interpretability of the models and conduct feature importance analysis to identify the factors that influence the predictions.

Conclusion

In conclusion, machine learning models based on ensemble methods can improve cardiovascular risk, despite this, data imbalance remains a significant challenge, making the application of oversampling techniques essential to ensure equitable distribution across classes. Integrating these models into clinical settings will require additional validation and adaptation to different populations. However, the growing application of AI in healthcare opens new opportunities for the development of personalized prediction systems, improving the early identification of at-risk patients, and contributing to the optimization of preventive strategies in the cardiovascular field.

Future research could focus on techniques such as SHAP or LIME to enhance the interpretability of the model and perform feature importance analysis to identify factors that influence the predictions. Furthermore, this work should increase the data taking into account diverse populations to assess their generalization and robustness.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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References

1. Lopez, E.O., Ballard, B.D. & Jan, A. Cardiovascular disease (2023).
2. Flora, G. D. & Nayak, M. K. A brief review of cardiovascular diseases, associated risk factors and current treatment regimes. *Curr. Pharm. Des.* **25**, 4063–4084. https://doi.org/10.2174/1381612825666190925163827 (2019).

3. Bays, H. E. et al. Ten things to know about ten cardiovascular disease risk factors – 2022. *Am. J. Prev. Cardiol.* **10**, 100342. https://doi.org/10.1016/j.ajpc.2022.100342 (2022).
4. Hajifathalian, K. et al. A novel risk score to predict cardiovascular disease risk in national populations (globorisk): A pooled analysis of prospective cohorts and health examination surveys. *Lancet Diabetes Endocrinol.* **3**, 339–355. https://doi.org/10.1016/S2213-8587(15)00081-9 (2015).
5. Mensah, G. A., Roth, G. A. & Fuster, V. The global burden of cardiovascular diseases and risk factors: 2020 and beyond. *J. Am. Coll. Cardiol.* **74**, 2529–2532. https://doi.org/10.1016/j.jacc.2019.10.009 (2019).
6. van Daalen, K. R. et al. Risk estimation for the primary prevention of cardiovascular disease: Considerations for appropriate risk prediction model selection. *Lancet Glob. Health* **12**, e1343–e1358. https://doi.org/10.1016/S2214-109X(24)00210-9 (2024).
7. Rojek, I. et al. Development of ai-based prediction of heart attack risk as an element of preventive medicine. *Electronics* https://doi.org/10.3390/electronics13020272 (2024).
8. Stonier, A. A., Gorantla, R. K. & Manoj, K. Cardiac disease risk prediction using machine learning algorithms. *Healthc. Technol. Lett.* **11**, 213–217. https://doi.org/10.1049/htl2.12053 (2024).
9. Li, J. et al. Comparison of the effects of imputation methods for missing data in predictive modelling of cohort study datasets. *BMC Med. Res. Methodol.* https://doi.org/10.1186/s12874-024-02173-x (2024).
10. Sozen, M. E., Sariyer, G., Sozen, M. Y., Badhotiya, G. K. & Vijavargy, L. Machine learning implementations for multi-class cardiovascular risk prediction in family health units. *Int. J. Math. Eng. Manag. Sci.* **8**, 1171–1187. https://doi.org/10.33889/IJMEMS.2023.8.6.066 (2023).
11. Jiang, Y. Cardiovascular disease prediction by machine learning algorithms based on cytokines in Kazakhs of China. *Clin. Epidemiol.* **13**, 417–428. https://doi.org/10.2147/CLEP.S313343 (2021).
12. Padhy, S. K., Mohapatra, A. & Patra, S. *Classification of Cardiovascular Disease Information System Using Machine Learning Approaches, 207–220* (Springer Nature Singapore, 2024).
13. Padhy, S. K., Mohapatra, A. & Patra, S. Enhanced hybrid framework for detection of cardiovascular disease based on adaptive neural network and fuzzy inference system. *Procedia Comput. Sci.* **258**, 2659–2670. https://doi.org/10.1016/j.procs.2025.04.527 (2025).
14. Padhy, S. K., Mohapatra, A. & Patra, S. We-xai: explainable ai for cvd prediction using weighted feature selection and ensemble classifiers. *Netw. Model. Anal. Health Inform. Bioinform.* **14**, 13. https://doi.org/10.1007/s13721-025-00508-2 (2025).
15. Zaidi, S. A. J. et al. Heartensemblenet: An innovative hybrid ensemble learning approach for cardiovascular risk prediction. *Healthcare (Switzerland)* https://doi.org/10.3390/healthcare13050507 (2025).
16. Blockeel, H., Devos, L., Frénay, B., Nanfack, G. & Nijssen, S. Decision trees: From efficient prediction to responsible ai. *Front. Artif. Intell.* **6**, 1124553. https://doi.org/10.3389/frai.2023.1124553 (2023).
17. Tandazo-Malla, C., Reátegui, R. & Suárez, R. Exploratory data analysis of cardiovascular disease data. In *8th International Conference on Information Technology & Systems*.
18. Chawla, N. V., Bowyer, K. W., Hall, L. O. & Kegelmeyer, W. P. SMOTE: Synthetic minority over-sampling technique. *J. Artif. Intell. Res.* **16**, 321–357. https://doi.org/10.1613/jair.953 (2002).
19. Hernandez, M. *et al.* Comparative evaluation of oversampling techniques for balancing metabolic profiles. In *Proc. of the 9th International Conference on Bioinformatics Research and Applications*, ICBRA '22, 41–47, https://doi.org/10.1145/3569192.3569200 (Association for Computing Machinery, New York, NY, USA, 2023).
20. Batista, G. E. A. P. A., Prati, R. C. & Monard, M. C. A study of the behavior of several methods for balancing machine learning training data. *SIGKDD Explor. Newsl.* **6**, 20–29. https://doi.org/10.1145/1007730.1007735 (2004).
21. Huang, W. et al. Application of ensemble machine learning algorithms on lifestyle factors and wearables for cardiovascular risk prediction. *Sci. Rep.* **12**, 1033. https://doi.org/10.1038/s41598-021-04649-y (2022).
22. Cai, Y. et al. Downscaling of smap soil moisture data by using a deep belief network. *Remote Sens.* **14**, 5681. https://doi.org/10.3390/rs14225681 (2022).
23. Bentéjac, C. et al. A comparative analysis of gradient boosting algorithms. *Artif. Intell. Rev.* **54**, 1937–1967. https://doi.org/10.1007/s10462-020-09896-5 (2021).

Author contributions

R.R Conceptualization, methodology, experimentation, validation, writing, C.T. conceptualization, data pre-processing, conclusion, R.S. conceptualization, data collection, L.R. conceptualization , writing—review.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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